

specifies, down to the syscalls, which capabilities each container is granted, ensuring it only gets what it genuinely needs to function.

5. Containers shouldn't be changing their state once they're up and running. We must enforce immutable infrastructure, meaning containers are rebuilt from scratch for every update rather than being patched live. This prevents attackers from persisting inside a compromised container. Think of this as Docker's "build once, deploy everywhere" mantra. It never truly worked for Java (also a technology that Linux

absolutely hates), but it might just work for a containeriser. Changes require redeployment, not modification, thus ensuring that every running instance is identical to the tested version.

6. Build in real-time vulnerability scanning and automated patching. Containers should be scanned

continuously, not just at build time. If a vulnerability is found, the system will either patch it in the background or alert you to rebuild the affected containers. This means integrating tools like Clair or Trivy directly into the platform, ensuring that no container runs outdated or vulnerable code. **END** 

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Container-native applications:

These applications are built specifically for containers, often following a microservices architecture. Spotify, for instance, uses container-native applications to break down its services into microservices, enabling faster development, deployment, and scaling. This approach not only improves flexibility and resource efficiency but also ensures that the apps can run seamlessly across multiple cloud environments, providing a more robust and scalable infrastructure.

Each of these trends shows how containerisation and orchestration are pushing the boundaries of modern software development, allowing companies to innovate, scale, and secure their applications more efficiently.

The future of containerisation and Kubernetes

Containerisation is changing quickly, and several key trends are shaping its future alongside platforms like Kubernetes.

Automation and AI: Automation and artificial intelligence (AI) are changing how we manage containers. Smart systems can automatically adjust resources based on real-time demand. For example, when a cricket match is

streaming, if there are many viewers, the system can allocate more resources to handle the traffic. Predictive maintenance tools can analyse system logs to predict failures. This means if a problem is detected, like a server about to crash, it can be fixed before it happens. Companies like Infosys use AI to monitor their systems, identify problems, and suggest solutions, reducing the need for human intervention.

Simplified developer experience:

Making it easier for developers to use container technology is essential. New, user-friendly interfaces are being created so developers can manage containers without needing deep knowledge of Kubernetes. Platforms like Google Cloud offer tools that allow even beginners to deploy applications with just a few clicks. Low-code and no-code solutions are helping non-technical users, like small business owners, quickly develop applications without writing a lot of code. Improved documentation and community support also help people learn and adopt these technologies.

Security enhancements: As more companies start using containers, security is becoming more important. Tools like Open Policy Agent help enforce security rules throughout the lifecycle of a container. For example, if a school is using a cloud-based platform to manage student data, they need to ensure that only authorised users can access it. Real-time monitoring can detect unusual activities, like unauthorised access attempts. Automated vulnerability scanning can identify security risks before deployment, ensuring applications are safe.

To conclude, while Docker and Kubernetes are widely used, there are alternatives like Podman and OpenShift that cater to different organisational needs. Looking ahead, trends such as automation and AI will further enhance container management, making it easier and more efficient for developers. The growth of edge computing and serverless architectures will also create new opportunities for organisations to innovate and stay competitive. **END** 

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